Guidance Note QGN 09: Reviewing the effectiveness of Safety and Health Management Systems

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Resources Safety & Health Queensland

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Reference is made to the following legislation as applicable to a Mine or Quarry in Queensland:

- *Coal Mining Safety and Health Act 1999*
- Coal Mining Safety and Health Regulation 2017

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1.0 Foreword

The *Coal Mining Safety and Health Act 1999* (CMSHA) and the *Mining and Quarrying Safety and Health Act 1999* (MQSHA) protect the safety and health of mine workers and those who are affected as a result of mining operations.

This guidance note was published in 2008 with references made to both CMSHA and MQSHA. Guidance for mineral mines and quarries has been separated from the guidance for coal mines and published as QGN38. This guidance note – QGN09 – is now specifically for coal mines in Queensland.

CMSHA and the Coal Mining Safety and Health Regulation 2017 (CMSHR) require that risks to safety and health from mining operations be identified and controlled to within acceptable limits and be as low as reasonably achievable (referred to as an acceptable level of risk). One of the primary ways this is achieved is that each mine develops a Safety and Health Management System (SHMS) that identifies hazards, examines attendant risk and ensures that these risks are controlled within acceptable limits. Under the legislation the Site Senior Executive (SSE) and the mine operator are given separate but complementary obligations to ensure that this is achieved.

The SSE has the obligation to develop and implement the mines' SHMS and to ensure the risk to persons from mining operations is at an acceptable level. The mine operator has the obligation to audit and review the system to ensure it is effectively controlling risks to within acceptable levels.

It is important for the operator to be aware that obligations cannot be transferred from the operator to the SSE in the discharge of this obligation. Attention is drawn to the provisions of the legislation that prevents a person with obligations from transferring the obligations to another person. The question has arisen on how to assess the effectiveness of an SHMS and this Guidance Note addresses this question.

The basis of the approach taken in the Guidance Note is to consider an SHMS as a dynamic system consisting of various subsystems each carrying out a vital function. Accepting this model if each subsystem is found to be working effectively then the overall system can be considered effective. This approach parallels the approach taken in diagnosing the condition of systems in science and engineering.

The obligation placed on mine operators regarding SHMSs is expressed in the terms 'audit' and 'review'. Auditing is simply the mechanism by which information is obtained for a particular purpose. For example, obtaining information to determine compliance (either with legislation or corporate standards) is a compliance audit; for the subject at hand obtaining information to determine effectiveness is an effectiveness audit.

It should be noted that most audits carried out in industry are compliance audits and the number of effectiveness audits carried out has been limited. The legislation does not define the word 'audit' however industry has accepted the meaning included in the various standards; namely to systematically examine documents and records to sufficient depth to verify whether the system meets some established criteria. Some definitions of auditing contain elements of review (see Definitions and Appendix A).

The term 'review' in this document is used in the sense that the information obtained by auditing is critically evaluated; in the Guidance Note this means evaluating whether the SHMS is keeping risk at an acceptable level.

Australian Standard: AS/NZS 4804 Occupational health and safety managements systems – General guidelines on principle, systems and supporting techniques clause 4.5 ‘Review and Improvement’ states the following “Management review is a cornerstone of the management system, providing an opportunity for senior management to regularly review the operation of the system and its continuing suitability in the face of change to make adjustments to build upon and improve its effectiveness”.

This Guidance Note endorses this statement.

2.0 Purpose and scope

This Guidance Note is provided to assist operators to meet their obligations under CMSHA and CMSHR to review the effectiveness and implementation of a mines’ SHMS to ensure the risk to persons from mining operations is at an acceptable level.

The document is not intended to be an exhaustive treatment of reviewing a SHMS but a guide. For example, two areas not specifically mentioned in this Guidance Note, which a concerned mine operator would review in detail, are compliance with legislation and the competence of mine workers to carry out the tasks. These areas are extensive enough to be worthy of separate reviews.

Any review carried out by an operator would have to take into account the conditions at a mine such as the complexity and associated hazard levels and past performance in establishing and maintaining an acceptable level or risk.

The Guidance Note identifies and examines some of the key subsystems that would be included in an effective SHMS if the system is to deliver an acceptable level of risk and remain capable of accommodating the changing circumstances that occur at every mine site.

The document is not a recognised standard as defined in CMSHA.

3.0 Introduction

An effective SHMS is a dynamic system that when implemented ensures that risks to the safety and health of mine workers are at an acceptable level and leads to continuously improving safety and health standards at the mine.

The SSE and the operator of a mine both have obligations regarding a mine’s SHMS. The SSE has an obligation to develop and implement a SHMS with the aim of ensuring the mine controls risks to an acceptable level. The mine operator has the obligation to review the SSE’s SHMS and determine whether it is implemented and working effectively and if necessary, require any a corrective action to be taken to make it effective.

This Guidance Note is designed to be a diagnostic tool to assist mine operators to meet their obligations to ensure the system is effectively controlling risk to an acceptable level.

The nature of a SHMS and its dynamic subsystems is illustrated in Figure 1 “Safety and Health Management System”. The operation of such a system and its subsystems is discussed and particularly how the various dynamic subsystems contribute to keeping risk within an acceptable level.

The system is examined with respect to the two requirements – to achieve a level of risk within acceptable limits and as low as reasonably achievable. Subsystems identified include:

- Change management
- Work force involvement
- System performance: lead and lag indicators
- Causal analysis: repairing defences
- Audit and inspection findings
- Contractor safety and health
- Chronic exposures causing incapacity

4.0 Safety and Health Management System

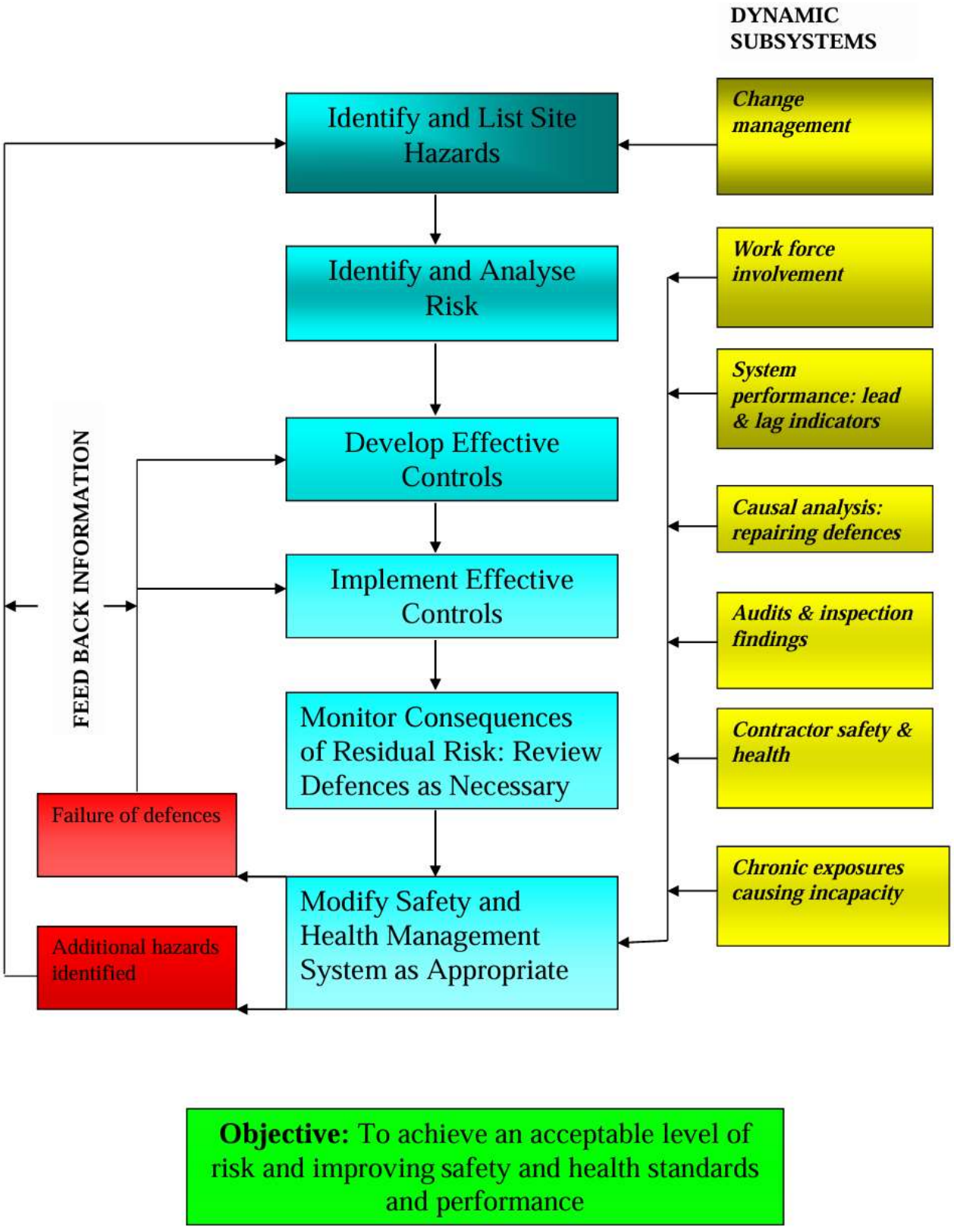


Figure 1: Safety and Health Management System

5.0 Obligations of operators and Site Senior Executives

Confusion has arisen over whether the obligation of an operator to review the effectiveness and implementation of a mine's SHMS can be discharged by the operator, delegating this duty to the SSE. This approach introduces doubt as to whether operators have indeed met their obligations under the legislation. The legislation quite specifically precludes the transfer of an obligation from one person to another.

It would be expected that once an SSE developed and implemented a SHMS at a mine, some form of auditing would be undertaken on behalf of the SSE to ensure the system is working as intended. However, any auditing controlled by the SSE would not be considered to meet the obligations placed on the operator to review the effectiveness and implementation of the system.

Doubt regarding the discharge of the operator's obligations will be greatly reduced, if not eliminated, if the review of a mine's SHMS is undertaken by suitably competent people, engaged by the operator, who are independent of the site senior executive and the mine being audited.

The intervals of the review should be determined by the results of previous reviews and recent safety performance of the mine. It is suggested that decisions on who should conduct the review, and at what intervals the reviews be carried out, are made at corporate level.

6.0 Acceptable level of risk

The reason a SHMS exists is to ensure the risk to mine workers safety and health is maintained at an acceptable level. A secondary but important objective is to generate ongoing improvement to safety and health standards at a mine.

To do this effectively a system must contain, in addition to the mechanisms to identify and analyse the risks and develop controls, adequate subsystems to detect weakness (accidents and high potential incidents etc) in existing controls and allow corrections. The subsystems must also allow for the monitoring of any site changes that affect risks and allow the development of appropriate controls to control these risks.

An acceptable level of risk is defined in CMSHA as within acceptable limits and as low as reasonably achievable. The requirement for the system to be dynamic cannot be over emphasised. An effective SHMS will soon become ineffective if the subsystems cannot detect and repair weaknesses to the existing system or accommodate site changes.

In summary, the subsystems labelled as dynamic in the diagram "Safety and Health Management System" have two important functions; they manage change and monitor the effectiveness of existing system.

The effectiveness of a SHMS is to a large extent dependent on the effectiveness of these dynamic subsystems and a review of a SHMS to determine effectiveness must closely examine these subsystems to ensure they are effective.

6.1 Risk within acceptable limits

To achieve a level of risk within acceptable limits a mine must have processes which form part of the SHMS to:

1. Identify hazards including principal hazards on site, identify and analyse any associated risks and develop and implement controls. In developing controls, the hierarchy of risk controls should be applied as appropriate to the level of risk determined.
2. Identify new hazards created on site as a result of change, identify and analyse any associated risks and develop and implement controls.
3. Monitor and record the consequences of residual risk.
4. Use the information obtained by 1 to 3 above to review the SHMS including, Principal Hazard Management Plans (PHMPs), standard operating procedures, standard work instructions, investigation techniques, risk management procedures etc and modify these as appropriate.

By their nature, principal hazards are associated with potential multiple fatalities, and care needs to be taken to ensure that they are properly audited and appropriately reviewed to ensure any audit recommendations are processed, i.e. adopted or rejected after analysis.

6.2 Risk as low as reasonably achievable

To ensure that risk is as low as reasonably achievable, the SHMS should include for the following:

1. Holding regular safety meetings to provide and obtain information from the work force on safety matters and minute proceedings and recommendations. Where a number of major contractors work on a site the safety and health information from these contractors needs to be co-ordinated. One method being used to achieve this requirement is for the SSE (or delegate) to attend regular meetings with contractor's safety and operational personnel to review contractor safety performance and ensure contractor's ongoing compliance with the mine's SHMS.
2. Ensuring that incidents and accidents are reported, monitored and properly analysed to identify immediate and underlying causes. An indication of an ineffective system is one in which incidents with the same underlying or root cause reoccur.
3. Monitoring and analysis of contractor safety and health performance. See section 10: Contractor Safety Performance, for this important aspect of safety and health management in the mining industry.
4. Carrying out regular safety observations, inspections and appropriate audits and establishing programs to address issues raised in these activities and monitor corrections.
5. Implementing requirements resulting from statutory inspections.
6. Using the information obtained from 1 to 5 above review the safety and health management system.

7.0 Workforce involvement

The workforce is the focus of the SHMS. The term is inclusive and involves all mine workers on site. As previously mentioned regular safety management meetings involving the integration of contractor's employees are a necessity if risk 'as low as reasonably achievable' is to be achieved.

However, workforce participation goes much further than regular safety meetings at which safety issues are raised, analysed and acted on. A mine should have a system that describes how workforce participation is achieved on site including the workforce of contractors.

An operator's review would assess the effectiveness of the system developed on site to involve the workforce.

8.0 System performance: lead and lag indicators

An effective SHMS should have built into the system indicators to allow the detection of system malfunctions and allow the ongoing performance of the system to be assessed. For this purpose, the system should include:

1. A suite of lead indicators.
2. A suit of performance indicators measuring safety and health standards and performance (lag indicators).
3. Provision for the regular monitoring of lead indicators and review of performance indicator trends.
4. Provision for the regular examination of the information obtained from 1 and 3 above to determine safety performance.
5. Provision for reviewing and modifying the system when safety performance is deteriorating or not improving.

An effective SHMS contains both lead and lag indicators; lead indicators indicate whether the system is working effectively; lag indicators provide information on the results being achieved by the system.

The Chamber of Minerals and Energy of Western Australian "Guide to Positive Performance Measurement in the Western Australian Mineral and Resources Industry" published in 2004 provides excellent information on developing lead indicators.

Lag indicators are important because, if recorded with integrity, they measure the success or otherwise of the SHMS in delivering and improving safety and health standards at the mine.

9.0 Causal analysis: repairing defences

The activities at a mine site involve people, materials and machines. It is the purpose of mine management to organise these factors of production in the most economically efficient manner to undertake the activities.

CMSHA and CMSHR places legal obligations on all people involved in mining activities to ensure that these activities are carried out with an acceptable level of risk. However, the SSE and the mine operator carry the principal obligation to ensure that a system is in place to control risk.

9.1 Reviewing the effectiveness of the Safety and Health Management System

The legislation requires that the management system monitor levels of risk and adverse consequences of retained residual risk.

The inevitable incidents that will occur are an indication that the level of residual risk may not be acceptable. Some of the defences may be less than ideal or become inappropriate as the task changes over time. An effective SHMS must be adjusting, modifying and creating new defences as analysis of accidents and incidents reveals that existing defences are inadequate.

There are a number of causal analysis tools available to determine both immediate and underlying causes of events occurring on site and identify any problems which need to be addressed. An effective SHMS will contain a subsystem to obtain this information and act on it. The operators review should assess whether this is being done and if so, is it being done in an effective manner?

It is expected an appropriate subsystem would deliver:

1. Factual reporting of all accidents and incidents at the mine.
2. Causal analysis of the events (accidents and incidents) identifying immediate and underlying causes including absent, failed or ineffective defences.
3. A system for reviewing underlying causes to detect any repetitious events or general failure types.
4. A process of ensuring the information derived from the subsystem is acted on to create, modify, or adjust defences.

10.0 Audit and inspection findings

Any review on behalf of the operator must examine previous audit, inspection and review findings to make sure that problem areas in the SHMS previously identified have been addressed.

Audits would include compliance audits against legislation and corporate standards; inspections would include inspections by the legislator as well as the results of internal inspections.

11.0 Contractor safety performance

With the extensive use of contractors within the mining industry it is particularly important that any review of the effectiveness a mine's SHMS examines the effectiveness of the system maintaining contractor safety and health standards.

The adoption by SSEs of the safety systems of large corporations working as contractors on site needs careful assessment. It is the SSE who has the obligation to develop and implement a SHMS for the mine controlling activities on site, including the activities of contractors.

The contractor's obligation is to ensure compliance with CMSHA, CMSHR and any applicable parts of the mine's SHMS.

When contractors bring safety and health procedures onto a mine site these procedures need to be mapped for consistency against any procedures existing at the mine before being adopted into the mine's safety and health management system. Particular attention needs to be given to the need for contractors to adhere to procedures developed on site in response to legislative requirements i.e. standard operating procedures and standard work instructions. This examination of the contractor's procedures should include measurement against any requirements in the legislation for the development of those procedures. This task should be completed before the contractor begins work on site and should be documented.

It is advisable to ensure that procedures across site are consistent, particularly when a number of contractors are working on the same site. Any review should detect and correct inconsistent safety procedures that may exist on site.

An operator's review should address the effectiveness of the subsystem for managing contractors. It would be expected that such a system would be extensive, commencing with principles for assessing suitable contractors, stating requirements for adoption of the mines' SHMS.

It is suggested that an operator's review of this subsystem would be assisted by the existence of a comprehensive report of the work undertaken to ensure that the mines' SHMS included the work of the contractor and the steps taken to ensure that this was fully understood by the contractor. It would be useful if the report also contained the process by which the ongoing adherence of the contractor with the mine's system was to be monitored. The more extensive the work undertaken at the mine by a contractor the more such a document would assist the operator's reviewing the effectiveness of control over contractor's activities.

12.0 Chronic exposures causing incapacity

Each year a number of persons leave the industry because of incapacity to carry on working in the industry. For a number of these people service in the industry has either been the cause of their medical condition or a major contributing factor towards it.

An effective SHMS should have provision to identify mine workers who fall into this category and to examine each of these cases to identify whether they are attributable to any chronic (long term) exposure to low level hazard.

This is an important issue as causes of the incapacity may be due to exposure to a chronic substandard condition such as whole of body vibration over a period of time, rather than a fall or other traumatic event.

As with all investigations, records of the results of the investigation and the causes of the injury or illness should be kept and reviewed so that the SHMS can be modified as a result of lessons learnt to prevent future occurrences.

13.0 Definitions

Audit	A systematic examination against defined criteria to determine whether activities and related results conform to planned arrangements and whether these arrangements are implemented effectively and are suitable to achieve the organisation's policies and objectives. Source AS/NZS 4801:2001 "Occupational health and safety management systems – Specification with guidance for use"
Review	In the context of this Guidance Note "review" means a critical re-examine of the system with the intention of determining whether the effectiveness of the SHMS.
Effective	In the context of this Guidance Note effective means that a safety and health management system reduces the level of risk to safety and health of persons affected by the operations of a mine to within acceptable limits and as low as reasonably achievable. It is considered that achieving this goal would result in continuous improvement of safety and health standards and performance.
Residual risk	Risk after controls have been implemented; reviews and causal analysis may reveal that residual risk is not at an acceptable level
Operator	Is the person or entity appointed as the operator by the holder of the mining tenure or when no other operator is appointed: Section 21 of CMSHA - Meaning of coal mine operator
Site Senior Executive	Is the person appointed by the operator as required by section 41(1) of CMSHA.

14.0 References

- *Coal Mining Safety and Health Act 1999*
- Coal Mining Safety and Health Regulation 2017
- AS/NZ 4801 Occupational health and safety management systems
- AS/NZS: 4804: Occupational health and safety management systems – General guidelines on principles, systems and supporting techniques.
- AS/NZS/ISO19011: 2003 Guidelines for quality and /or environmental management systems auditing
- AS/NZS 4360: 2004 Risk Management
- Guide to Positive Performance Measurement in the Western Australian Minerals and Resources Industry

15.0 Appendix A: Notes on audits

Australian Standard AS/NZS/ISO19011: 2003 Guidelines for quality and/or environmental management systems auditing, states that an audit is a “systematic, independent and documented processes for obtaining audit evidence and evaluating it objectively to determine the extent to which the audit criteria are fulfilled.” where:

- “audit evidence” means “records, statements of fact or other information, which are relevant to the audit criteria and verifiable, and
- “audit criteria” means “sets of policies, procedures or requirements.”

A similar definition of an audit is given in AS/NZS 4801: 2001 Occupational health and safety management systems – Specification with guidance for use, and is “A systematic examination against defined criteria to determine whether activities and related results conform to planned arrangements and whether these arrangements are implemented effectively and are suitable to achieve the organization's policy and objectives.”

Whatever definition of an audit a mine decides to adopt, the following guidance is given so that an audit represents the current state of affairs and is a useful tool for confirmation of best practice and action for improvement.

The audit criteria, (defined criteria in AS/NZS 4801), should always include the legislative compliance requirements of a SHMS as described in CMSHA.

In addition, there are four key aspects to an audit on a SHMS and all aspects should be included:

1. Determine how the SHMS is intended to ensure the risks to persons from operations at the mine are at an acceptable level.
2. Establish whether the SHMS is implemented and effective in ensuring the risks to persons from operations at the mine are at an acceptable level.
3. Examine whether the SHMS is suitable for ensuring the risks to persons from operations at the mine are at an acceptable level.
4. Use of an evidence-based approach where audit evidence is verifiable. The audit should be based on samples of the information available.

Persons competent in auditing protocols and familiar with the relevant AS/ANZ and ISO standards should carry out the audit. An audit report and conclusion on the SHMS should include evidence of how well the planned arrangements have been implemented, how effective they are and how suitable they are.

The results of the audit will contain information on where risks are being well managed and identify opportunities for improvement. The auditing process should include a method for identifying and making improvements, ideally with linkage back to the mine’s SHMS.